

Problem Statement Report

Smart Agriculture Advisory System for Small and Marginal Farmers

1. Problem Overview

1.1 Background

India has over 140 million farm households, of which 86% are small or marginal farmers cultivating less than 2 hectares of land (NABARD Report, 2022). These farmers form the backbone of Indian agriculture, yet they remain the most vulnerable to knowledge gaps, unpredictable weather, pest attacks, and volatile market prices.

The majority of small farmers in India continue to rely on traditional knowledge, local input dealers, or informal community networks for crop selection, pest management, and fertilizer use. Access to scientific, personalised advisory remains largely limited to large commercial farms and government extension workers — a system stretched too thin to reach every village.

1.2 Core Problems

- ❶ No personalised crop advisory based on individual soil type, microclimate, or crop history.
- ❷ Heavy reliance on overuse of pesticides and fertilizers, causing soil degradation and rising input costs.
- ❸ Language barriers and low digital literacy prevent adoption of existing agri-tech apps.
- ❹ No real-time market price visibility, forcing farmers to accept distress prices from middlemen.
- ❺ Pest and disease identification relies on guess-work until significant damage has already occurred.
- ❻ Weather alerts are not localised or delivered in accessible formats (voice, local language).

1.3 Impact Data

Statistic	Source
86% of Indian farmers are small/marginal	NABARD Report, 2022
ICT-based advisories can increase crop yield by 20–30%	Ministry of Agriculture Studies
Farmers lose ~₹9,000 crore annually due to improper pesticide use	Indian Council of Agricultural Research (ICAR)

Only 1 extension officer available per 1,000 farmers in India

OECD Agricultural Outlook, 2023

2. Existing Solutions & Limitations

2.1 Current Offerings in the Market

Solution	Features	Strengths	Limitations
Kisan Suvidha (Govt)	Weather, Market prices, Agri advisories	Free, Government-backed	Poor UX, English-heavy, no AI personalisation
IFFCO Kisan App	Advisory, Weather, Mandi prices	Multi-language support	No pest image detection, requires good internet
AgroStar	Pest ID, input procurement, agri feed	Good pest detection module	Subscription model, urban-biased, not for marginal farmers
Plantix (PEAT)	AI plant disease diagnosis via photo	Strong CV model	No crop recommendation or price tracking, limited Indian crops
eNAM Portal	Digital mandi platform & prices	Official government market data	No mobile-first advisory, no AI layer, no voice

2.2 Key Gap our Solution Addresses

No existing solution combines all five required capabilities — crop advisory, pest detection, market prices, weather alerts, and multilingual voice support — into a single offline-capable, low-literacy-friendly mobile app targeted specifically at small and marginal farmers in India. Our proposed system fills this gap with an AI-first, modular architecture.

3. Proposed Solution

3.1 Solution Overview

KrishiMitra ("Farmer's Companion") is an AI-powered, multilingual mobile advisory platform for small and marginal farmers. It delivers real-time, personalised crop guidance through five independent but integrated modules, accessible via text, image, and voice in the farmer's native language — even on low-end smartphones with intermittent connectivity.

3.2 Module Architecture

Module 1 — Crop Recommendation Engine

Farmers input (or select via voice) their soil type, location, and season. The ML model processes N/P/K ratio, soil pH, local rainfall, temperature, and humidity to recommend the top 3 crops with expected yield range and fertilizer guidance. Model: XGBoost / Random Forest trained on the Atharva Ingle Crop Recommendation Dataset (22 crops, sourced from ICFA).

Module 2 — AI Pest & Disease Detection

Farmers upload a photo of the affected crop leaf. An EfficientNet-B0 model (transfer-learned on PlantVillage — 54,306 images, 38 disease classes) identifies the disease with >93% accuracy and returns the disease name, severity, and recommended treatment in plain language. The model is exported to TensorFlow Lite for on-device inference — no internet required.

Module 3 — Weather Alerts & Predictive Insights

Location-based weather forecasts (Open-Meteo API — no key required, 1–11 km resolution) generate smart alerts such as 'High fungal risk tomorrow due to humidity' or 'Avoid spraying — rain in 6 hours'. IMD Meghdoot advisories (official government source, 11 regional languages) are also integrated for district-level agri bulletins.

Module 4 — Market Price Tracker

Live mandi prices for the farmer's commodity and nearest regulated markets are fetched from data.gov.in (Agmarknet API — free, covers 3,000+ mandis, 200+ commodities). An LSTM model provides 7-day price trend forecasts and recommends the optimal market to sell in. Alerts notify farmers when prices rise above their set target.

Module 5 — Multilingual Voice Interface

The app supports Hindi, Punjabi, Tamil, Telugu, Marathi, Bengali, Gujarati and Odia via AI4Bharat's IndicTrans2 translation engine. A voice-first mode (text-to-speech + speech-to-text) allows fully illiterate farmers to navigate and receive advice by speaking in their local dialect. All advisory output is also rendered as short audio clips.

3.3 Unique Value Propositions

- ⚡ Offline-capable pest detection — TFLite model runs entirely on-device; no internet needed.
- 🗣️ Voice-first for low-literacy farmers — full app navigable by speech in 8+ Indian languages.

-  Unified platform — only solution combining all 5 advisory functions in a single free app.
-  Government data sources — Agmarknet + IMD data builds trust with farmer communities.
-  Modular architecture — each feature independently demonstrable and testable.

4. Data Sources

All datasets and APIs used in this project are publicly available at zero cost for non-commercial and academic use.

Dataset / API	Module	Content	Source URL
PlantVillage Dataset	Pest Detection	54,306 leaf images, 38 disease classes, 14 crops	huggingface.co/datasets/mohanty/PlantVillage
PlantDoc Dataset	Pest Detection	Real-field images for better generalisation	github.com/pratikkayal/PlantDoc-Dataset
Crop Recommendation Dataset	Crop Advisory	N, P, K, pH, temp, humidity, rainfall — 22 crops	kaggle.com/datasets/atharvaingle/crop-recommendation-dataset
Crop Yield (Indian States)	Crop Advisory	State-wise historical crop yield data	kaggle.com/datasets/akshatgupta7/crop-yield-in-indian-states-dataset
Agmarknet — data.gov.in	Market Prices	Daily min/max/modal mandi prices, 3,000+ mandis	data.gov.in — Agmarknet catalogue
Indian Mandi Prices 2023–2025	Price Forecasting	Cleaned historical mandi price dataset	kaggle.com/datasets/arjunyadav99/indian-agricultural-mandi-prices-20232025
Open-Meteo API	Weather Alerts	Hourly/daily forecasts, 1–11 km resolution, no API key	open-meteo.com
IMD Meghdoot Advisories	Weather Alerts	District agri advisories in 11 Indian languages	meghdoot.imd.gov.in
ISRO Bhuvan Soil Maps	Crop Advisory	India-wide soil type, pH, texture data	bhuvan.nrsc.gov.in
IndicTrans2 / IndicBERT	Multilingual NLP	Translation & NLP for 22 Indian languages	github.com/AI4Bharat/IndicTrans2

5. Technology Stack

Layer	Technology	Justification
Mobile Frontend	React Native + Expo SDK	Single codebase for Android + iOS; critical for rural low-end devices
UI Styling	NativeWind (Tailwind for RN)	Rapid UI development; consistent design system
Backend API	Python FastAPI	High performance, async, auto Swagger docs
Database	PostgreSQL + Redis	Relational store + cache layer for weather/price lookups
ML — Pest CV	EfficientNet-B0 → TFLite	Transfer learning on PlantVillage; on-device inference = offline capable
ML — Crop Rec	XGBoost / Random Forest	Tabular classification; fast training + interpretable results
ML — Price Forecast	LSTM (PyTorch)	Time-series prediction on historical mandi prices
Multilingual NLP	AI4Bharat IndicTrans2	SOTA translation for 22 Indian languages, open source
Voice I/O	Expo Speech + React Native Voice	Voice-first UX for low-literacy farmers
Weather API	Open-Meteo (no key required)	Free, no rate limits, 1–11 km resolution globally
Market Data API	data.gov.in / Agmarknet API	Official GoI data; free after registration
Push Notifications	Firebase Cloud Messaging	Free tier, reliable delivery for weather + price alerts
Deployment	Docker + Render (free tier)	Containerised, CI/CD via GitHub Actions

6. System Flow

6.1 End-to-End User Journey

Step 1 — Onboarding: Farmer opens app → selects language (voice or tap) → enters village/district location, primary crop(s), and soil type (assisted by dropdown + voice).

Step 2 — Crop Advisory: Backend queries Crop Recommendation model with N/P/K + weather context → returns top 3 recommended crops with yield estimate, fertilizer schedule, and sowing calendar in local language.

Step 3 — Pest Detection: Farmer taps camera icon → captures leaf image → EfficientNet TFLite model runs on-device → disease identified in <2 seconds → remedy displayed as text + audio in local language.

Step 4 — Weather Alerts: Background job pulls Open-Meteo forecast every 6 hours → rule engine checks for adverse conditions (rain, high humidity, frost) → push notification sent via FCM in local language.

Step 5 — Market Prices: Farmer selects commodity → app fetches live Agmarknet prices for 5 nearest mandis → LSTM model projects 7-day price trend → app highlights best market to sell with expected price.

Step 6 — Feedback Loop: App collects anonymous usage events and farmer-rated helpfulness scores → stored in PostgreSQL → used to retrain models monthly.

6.2 Data Flow Diagram (Text Representation)

Input / Trigger	System Response
Farmer photo of diseased leaf	On-device TFLite → disease name + remedy
Soil type + location entered	FastAPI → XGBoost model → top 3 crops + fertilizer
Weather service (every 6 hrs)	Alert engine → FCM push → farmer phone
Commodity selected by farmer	Agmarknet API → LSTM forecast → best mandi recommendation
Voice input in Hindi / regional	IndicTrans2 → intent extraction → module routing → voice response

7. Expected Outcomes & Impact

7.1 Functional Deliverables

- Fully functional Android + iOS mobile app (React Native / Expo) with 5 integrated modules.
- Pest detection model (EfficientNet-B0) trained on PlantVillage — target accuracy >93%.
- Crop recommendation model (XGBoost) — target accuracy >90% on held-out test set.
- LSTM price forecasting model — 7-day mandi price prediction with MAPE < 8%.
- Multilingual support for 8 Indian languages via IndicTrans2.
- Full voice navigation for zero-literacy users.

7.2 Impact Metrics

Expected Impact	Evidence Basis
20–30% increase in crop yield for adopting farmers	Ministry of Agriculture ICT studies
15–25% reduction in pesticide overuse	ICAR precision agri research
Better mandi price realisation — reduce distress selling	eNAM price transparency impact reports
Accessible to 86% of Indian farmers (small/marginal)	NABARD 2022

7.3 Beneficiaries

- Primary: Small and marginal farmers (<2 ha landholding)
- Secondary: Agricultural extension officers, NGOs, cooperative societies
- Tertiary: Government agriculture departments for data-driven policy decisions

8. Team & Implementation Plan

8.1 Sprint Plan (36 Hours — Hackathon)

Sprint	Tasks	Hours
1	Backend setup (FastAPI), DB schema, API integrations (Open-Meteo, Agmarknet)	0–8 hrs
2	Train & export Crop Recommendation model (XGBoost) + REST endpoint	4–12 hrs
3	Mobile app scaffold (React Native), onboarding flow, language selection	4–14 hrs
4	Pest Detection module — load PlantVillage TFLite model, camera integration	10–20 hrs
5	Market price screen + LSTM model, push notification setup (FCM)	16–26 hrs
6	Voice interface (IndicTrans2 + Expo Speech), multilingual UI	22–32 hrs
7	Integration testing, demo preparation, presentation slides	32–36 hrs

9. Conclusion

KrishiMitra addresses a genuine and urgent gap in Indian agriculture by placing AI-powered advisory directly in the hands of 140 million small and marginal farmers — in

their language, on their phone, even without internet. By combining four open-source ML models with three free government APIs and an offline-capable architecture, the solution is both technically feasible to build within 36 hours and sustainable to scale post-hackathon.

The modular design ensures every sub-system is independently valuable and demonstrable. The use of IMD and Agmarknet (official GoI data sources) adds credibility and alignment with the Ministry of Agriculture's digital agriculture mission. Voice-first design and support for 8 Indian languages ensure the solution reaches its intended beneficiaries rather than remaining accessible only to digitally literate users.

This project directly addresses SIH 2025 Problem #25010 and supports the national goal of doubling farmers' income through technology-driven precision agriculture.